

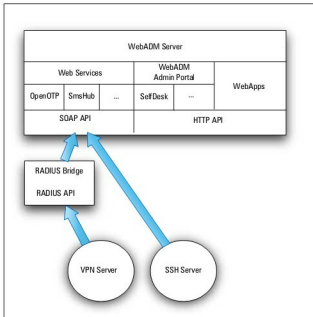


Figure 1: The suite components

scalable and highly available. After a bit of searching, I found a couple of enterprise products based on Linux and open source as candidates. After some testing, I approved of *rcdev's* product, OpenOTP.



Note: Briefly, RADIUS (Remote Authentication Dial In User Service) is a client-server protocol for AAA (authentication, authorisation and accounting) for applications such as network access or IP mobility. Intended to work in both local and roaming situations, it allows a company to maintain centralised user profiles that all remote servers can share. The RADIUS API interfaces with VPN servers and many other software applications requiring end-user authentication. For more information, please refer to en.wikipedia.org/wiki/RADIUS.

An advantage of OpenOTP is that we can use cell phones as our token device, which reduces the solution cost, since we don't need to buy hardware tokens (though this product supports those as well). Also, it's certified on the Open Authentication Framework (OATH). OpenOTP supports OATH HOTP (event-based), TOTP (time-based) and OCRA (challenge-based) standards for both software and hardware tokens. Many OATH-compliant software tokens are available for iPhone, Android, J2ME mobile phones, Windows Mobile, Palm, BlackBerry, etc.

Overview

OpenOTP is an enterprise-grade two-factor authentication solution based on open standards and technologies. It provides multiple authentication methods for LDAP users, which rely on OTP, including Mobile-OTP software tokens, SMS OTP, Secure Mail OTP and YubiKey. It is completely

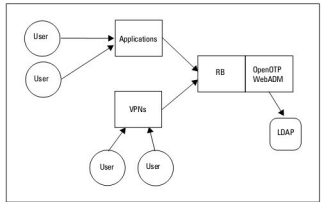


Figure 2: Integrating OpenOTP in your environment

free for up to 25 users. If you need more users, then you have to buy a licence. To learn more, please visit the website <http://www.rcdevs.com/>.

The OpenOTP solution is composed of the WebADM server application, the OpenOTP SOAP service, the optional RADIUS Bridge, User Self-service Desk, and the end-user Web application. The main components in the solution are:

- WebADM, Web services (e.g., OpenOTP), Web apps (e.g., SelfService)
- The LDAP server
- RADIUS Bridge
- PAM Bridge

The WebADM server contains:

- The SOAP APIs for the installed Web services
- The Web apps
- The admin portal
- The PKI server
- The session server

This article will not cover all these components; for a better idea of each component, please have a look at the product website. We will cover installation and configuration of OpenOTP, and how to configure some applications like SSH and RADIUS Bridge. Figure 1 shows the interaction of the components.

The current version of WebADM supports any 32-bit or 64-bit Linux with GLIBC >=2.5 and installed 32-bit binaries. The installation package contains the required dependencies, so you don't need to install anything extra. However, WebADM needs an LDAP server like OpenLDAP, and for logging, an SQL database (MySQL or PostgreSQL). I am using OpenLDAP and MySQL. This product can run both on a physical machine, or on a virtual machine. For the hardware requirements, please check the product website.

Figure 2 shows (at a high level) an overview of how OpenOTP might be integrated into your infrastructure.

Installing the product

Download the tarball from www.rcdevs.com/downloads/, uncompress it, and run it, as follows: